Technical University of Crete
School of Electrical and Computer Engineering
Course: Reinforcement Learning and Dynamic Optimization
Programming Assignment 2 : Reverse Engineering Non-Stationarity and Price Wars
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Instructor: Thrasyvoulos Spyropoulos

Areti Xynompiti

1 Overview of Part A

In this part of the assignment, we study a non-stationary two-armed bandit environment, where at each time step, the learner chooses one of the two arms and receives a Bernoulli reward. The environment is non-stationary because the reward probability of each arm changes periodically over time between a low and a high value, which we have the power to tune in order to reveal the weaknesses of the following algorithms.

2 Implementation of algorithms

2.1 Choice of algorithm parameters

2.1.1 Choice of Exploration Horizon

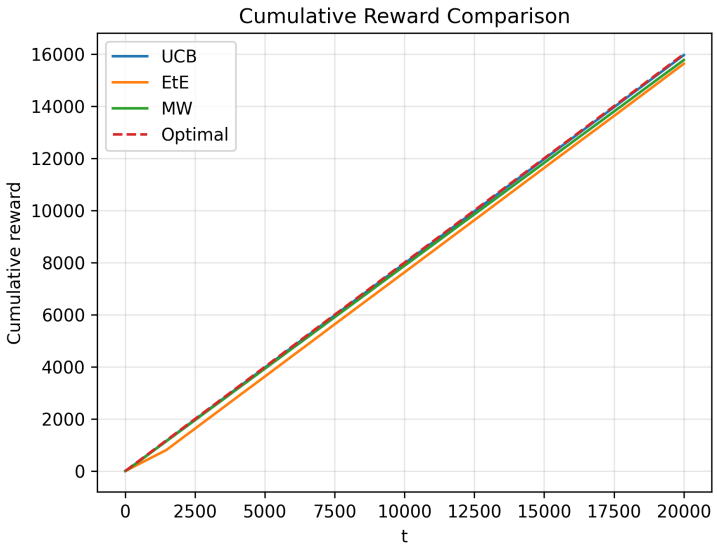
For Explore-then-Exploit, the exploration length N is chosen by balancing two opposing effects in the regret bound $\mathbb{E}[R(T)] \leq NK + 2T\sqrt{\frac{2\log T}{N}}$. The first term is the exploration cost, which increases with N, while the second term is the exploitation error, which decreases with N, since larger exploration improves the accuracy of the empirical mean estimates. Balancing these two contributions yields : $N = O\Big(\big(\frac{T}{K}\big)^{2/3}(\log T)^{1/3}\Big)$, and by ignoring K and the log term we end up using: $N_{\text{explore}} = T^{2/3}$.

2.1.2 Choice of MW parameters

For MW in bandit mode, the choice is based on the regret bound: $L_{\text{ALG}} \leq (1+\eta)\min_i \sum_{t=1}^{T} \ell_i^{(t)} + \frac{K\ln K}{\epsilon\eta} + \epsilon T$. In the last two terms we can see the tradeoff in the choice of ϵ and η: if they are too small, the term $\frac{K\ln K}{\epsilon\eta}$ becomes large, while if ϵ is too large then the exploration cost ϵT dominates. Choosing ϵ and η on the same scale, $\epsilon \approx \eta \approx x$, reduces the tradeoff to $\frac{K\ln K}{x^2} + xT$ and by balancing these two terms we obtain $x \approx \left(\frac{K\ln K}{T}\right)^{1/3}$

Therefore, in our implementation we use $\eta = \left(\frac{K\ln K}{T}\right)^{1/3}, \epsilon = \left(\frac{K\ln K}{T}\right)^{1/3}$

2.2 Stationary Check



	Arm 1	Arm 2
p^L	0.8	0.3
p^H	0.8	0.3
T^L	700	700
T^H	700	700
Initial state	H	L

Here, although the environment alternates between periods, it is actually stationary, since for each arm the low and high probabilities are the same. So arm 1 is always better than arm 2, and we expect all algorithms to eventually learn to mostly choose it,which is exactly what we observe in the plot.

3 Scenario designed so MW beats EtE

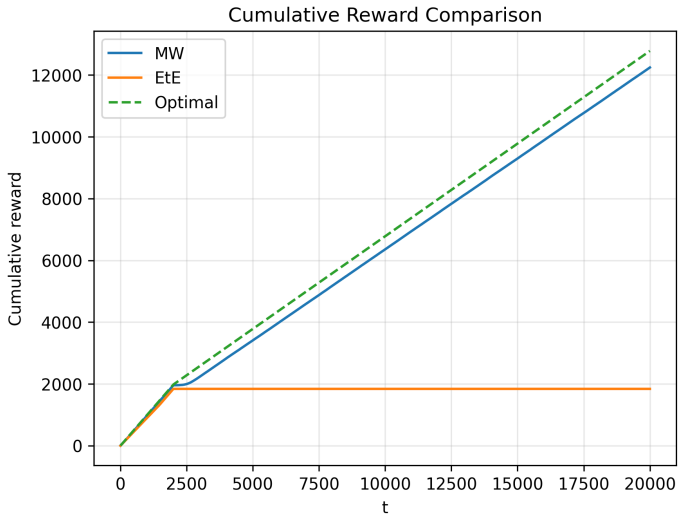
3.1 Environment Parameters

To construct a scenario where MW beats EtE, we first think about what hurts EtE the most. Since it explores only during the first $2 \cdot N_{\text{explore}}$ rounds and then commits to the arm that looked best, we can make one arm look good in the exploration phase and bad later, so that EtE gets trapped by this early estimate and cannot adapt properly after the change and this is exactly what we do by choosing the follwing parameters.

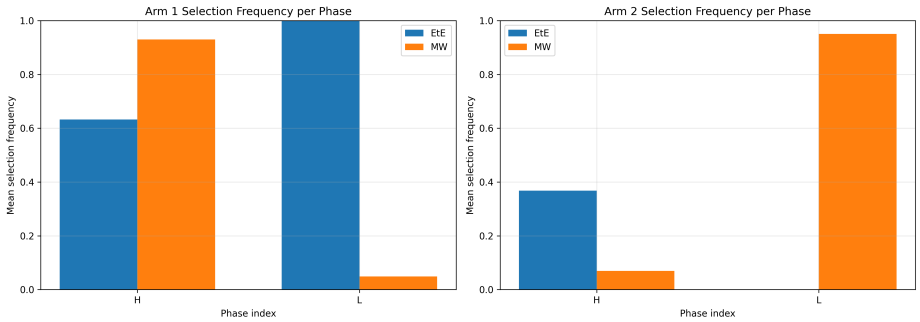
	Arm 1	**Arm 2**
p^L	0.0	0.6
p^H	0.99	0.8
T^L	18000	18000
T^H	2000	2000
Initial state	H	H

3.2 Results

EtE completely collapses in this scenario as we can see in the cumulative reward plot, after the exploration phase its curve becomes flat, which shows that it has committed to the wrong arm and then keeps receiving zero reward. Also in the arm-selection plots,in the low phase EtE keeps selecting arm 1 all the time, even though arm 2 is now the better arm, while MW moves almost entirely toward arm 2.



Algorithm	**Mean ± Std**
EtE	1839.050 ± 13.970
MW	12243.750 ± 128.170



4 Scenario designed so MW beats UCB

4.1 Environment Parameters

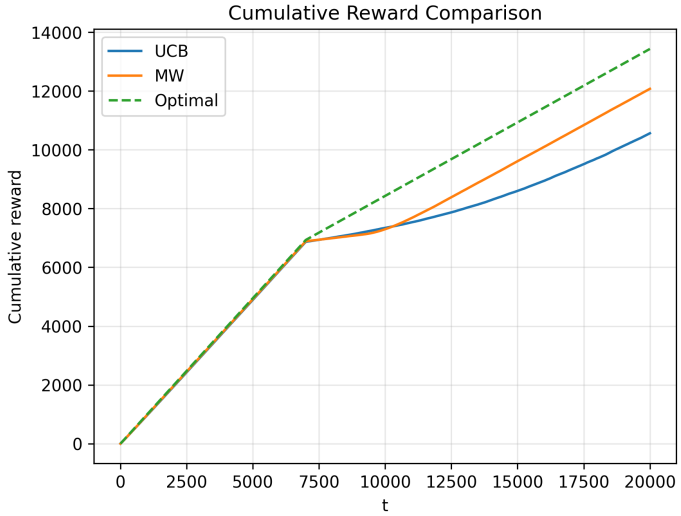
The main idea behind this parameter choice was to first let the algorithm play the initially better arm and then switch do an environment where the optimal arm has chnaged. Before the switch we want $N_{\text{optimal}} \gg N_{\text{suboptimal}}$, because at that point the empirical mean of the optimal arm is high and the exploration term is very low, therefore

the empirical mean becomes the dominant part of its UCB value.
We also want this switch to happen at a point where the above is guaranteed, but its's still early enough so that there is enough horizon left afterwards for UCB to suffer regret. In this way, when the environment changes and let's say arm 2 becomes the better arm, UCB still relies heavily on the old estimate of arm 1 ,so in order for UCB to adapt, this estimate must decrease,which is something that happens very slowly.

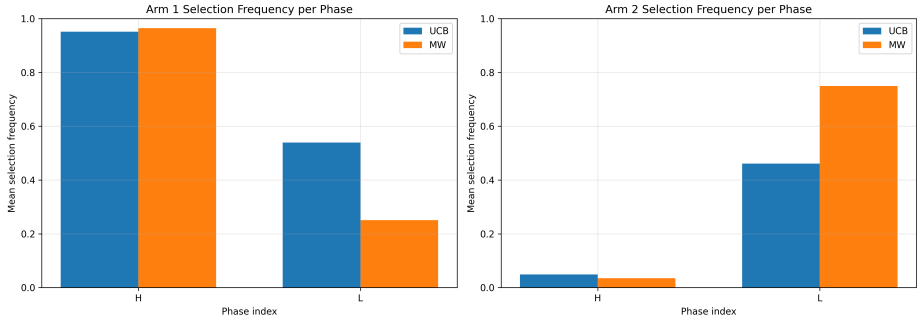
	Arm 1	**Arm 2**
p^L	0.1	0.5
p^H	0.99	0.8
T^L	13000	13000
T^H	7000	7000
Initial state	H	H

4.2 Results

After the change point, UCB still keeps choosing arm 1 for a long time, even though arm 2 is now the truly better arm. This can be observed in the cumulative reward plot, where after the switch the UCB curve starts falling behind MW. The reason is, as we mentioned previously, that arm 1 has already been played many times in the first phase, so its empirical mean is still very high and does not decrease quickly. In other words, for arm 2 to take the lead, the UCB value of arm 1 must first decrease enough so that arm 2 can take the lead consistently, and not just temporarily because of exploration.
This behavior is also reflected in the arm-selection plots, since in the high phase both algorithms mostly choose arm 1, while in the low phase UCB still selects arm 1 much more often than MW, which moves more rapidly toward arm 2.
Eventually UCB does start moving toward the correct arm, but this happens quite late, after it has already wasted many rounds on arm 1, and this is exactly what creates the gap in cumulative reward. The fact that UCB under-performs can also be veryfied from the table of mean rewards, since the mean reward of MW is much higher than that of UCB, and this difference is much larger than the corresponding standard deviations, so the gap is not just due to randomness.



Algorithm	**Mean ± Std**
UCB	10560.200 ± 140.843
MW	12072.900 ± 289.766



5 Overview Part B

In Part B, we study the pricing game in the case where the competitor reacts strategically to the learner's action and we examine this setting under two learning algorithms.
In the UCB case, the learner chooses a single price at each round and the competitor can directly react to that value. In the MW case, the learner plays according to a mixed distribution over the available prices, so the competitor responds to the corresponding probability distribution and not one deterministic price.

6 Competitor's myopic best response

6.1 Best response against UCB

At each round t, the learner first selects its price p_t and since the competitor observes this value before making its own decision, p_t is known and treated as a constant when the competitor chooses q_t. Therefore, the competitor's problem reduces to maximizing: $f(q) = \frac{q}{1+e^{\beta(q-p_t)}}$.
We differentiate with respect to q and set the derivative equal to zero, since only the numerator of the derivative has to be zero , we obtain the nonlinear equation $h(q) = 1 + e^{\beta(q-p_t)} - \beta q e^{\beta(q-p_t)} = 0$. This equation does not admit a simple closed-form solution, so in the implementation we solve it numerically using the bisection method. We choose this method because $h(q)$ is continuous in q, and we can find an interval $[a, b]$ such that

$$h(a) > 0, \qquad h(b) < 0.$$

Hence, by Bolzano's theorem, there exists at least one root in $[a, b]$, so the bisection method can be applied. In the implementation, we start from $a = 0$ and choose b large enough until $h(b) < 0$, then bisection converges to a solution of $h(q) = 0$.

6.2 Best response against MW

In the MW setting, the learner does not choose a price deterministically. Instead, at round t, it samples an arm according to the mixed distribution

$$q_i^{(t)} = (1-\epsilon)p_i^{(t)} + \frac{\epsilon}{K}, \text{where } p_i^{(t)} \text{ is the distribution induced by the current weights.}$$

Thus, if P_t denotes the learner's random price, then $\mathbb{P}(P_t = p_i) = q_i^{(t)}, \qquad i = 1, \ldots, K.$
For a fixed competitor price q, the probability that the competitor wins against learner price p_i is

$$\mathbb{P}(\text{competitor wins} \mid P_t = p_i) = \frac{1}{1 + e^{\beta(q-p_i)}}.$$

Since the competitor earns q when it wins and 0 otherwise, its conditional expected reward is:

$$\mathbb{E}[R_t^{\text{comp}} \mid P_t = p_i] = q \cdot \frac{1}{1 + e^{\beta(q-p_i)}}.$$

By taking expectation over the learner's mixed distribution, we obtain:

$$\mathbb{E}[R_t^{\text{comp}}] = \sum_{i=1}^{K} \mathbb{P}(P_t = p_i)\, \mathbb{E}[R_t^{\text{comp}} \mid P_t = p_i] = \sum_{i=1}^{K} q_i^{(t)} \frac{q}{1 + e^{\beta(q-p_i)}}.$$

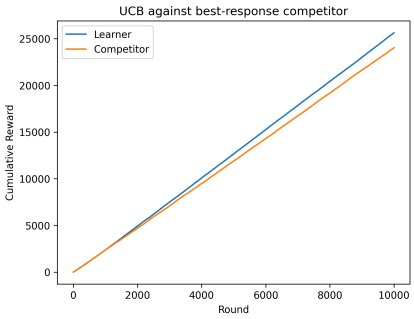
Therefore, the competitor chooses its price by solving

$$\max_{q \geq 0} \sum_{i=1}^{K} q_i^{(t)} \frac{q}{1 + e^{\beta(q-p_i)}} = \max_{q \geq 0} R_t(q)$$

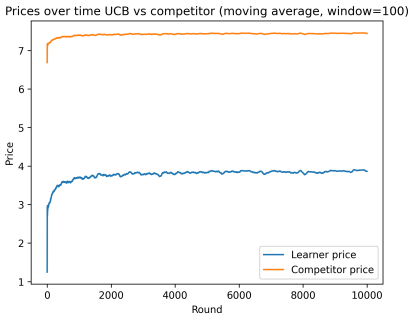
Because the problem is one-dimensional and differentiable, we can compute the best response numerically by solving the first-order condition $R_t'(q) = 0$ via bisection method as before.

7 Experiments for different price sensitivities

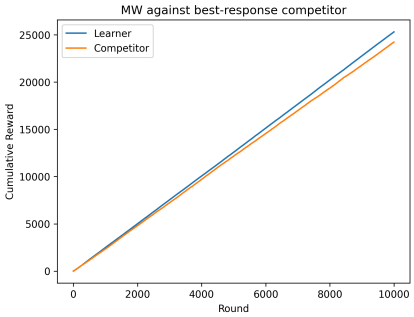
7.1 Low price sensitivity



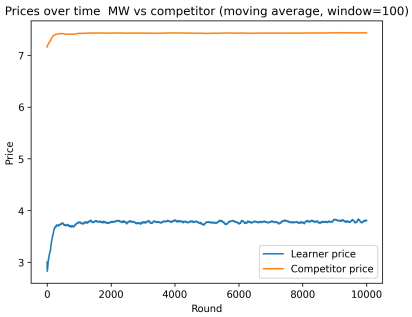
(a) UCB cumulative reward



(b) UCB prices over time



(c) MW cumulative reward

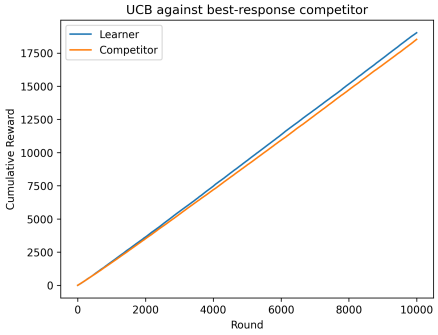


(d) MW prices over time

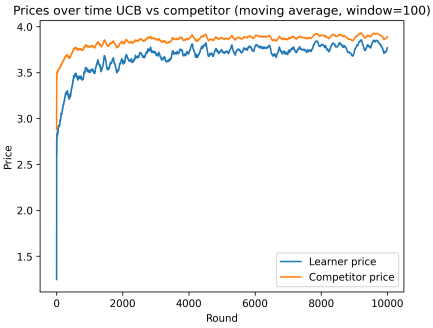
At $\beta = 0.2$, although the learner remains at lower price levels, it keeps a higher probability of being chosen as the cheaper seller. In this regime, that winning-probability advantage is sufficient to compensate for the lower reward obtained from each individual sale.
The two algorithms seem to perform very similarly in this environment, since the difference in their mean rewards is not large compared to the relatively high standard deviations they experience.

UCB Learner: 25840.250 ± 169.665
MW Learner: 25306.966 ± 224.391

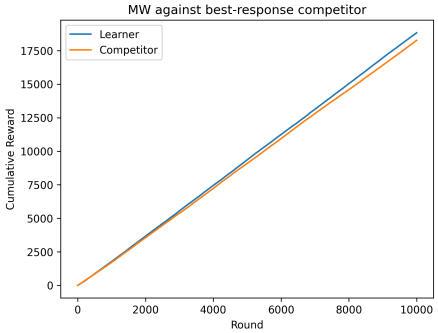
7.2 Medium price sensitivity



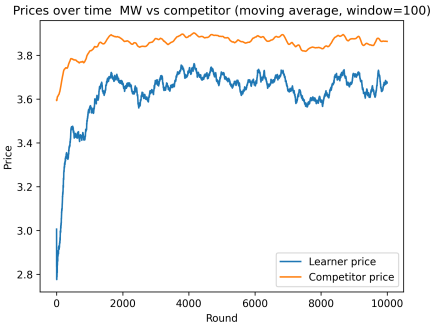
(a) UCB cumulative reward



(b) UCB prices over time



(c) MW cumulative reward

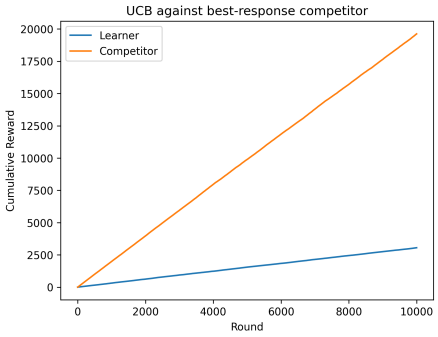


(d) MW prices over time

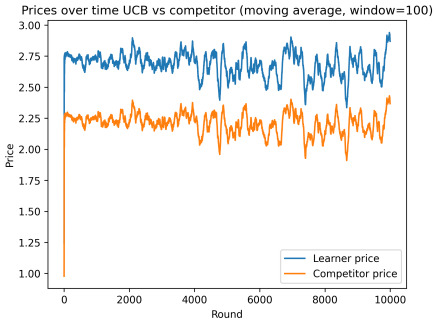
At $\beta = 0.5$, the learner is again typically priced below the competitor, so both learners still keep a winning-probability advantage and remain above the competitor in cumulative reward. However, customer choice is now more sensitive to price differences than at $\beta = 0.2$. As a result UCB performs slightly better than MW in this regime. In the UCB case, the learner keeps prices that are higher than MW's, while still staying clearly below the competitor. Therefore, UCB maintains the learner's winning-probability advantage and at the same time obtains more reward per successful sale than MW.

UCB Learner: 19361.644 ± 149.425
MW Learner: 18831.243 ± 225.200

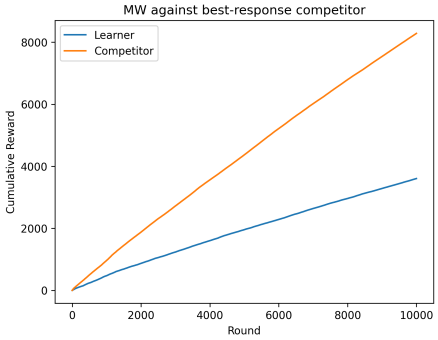
7.3 High price sensitivity



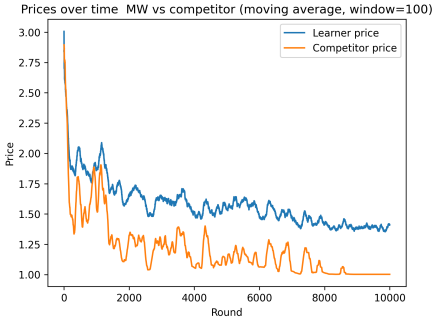
(a) UCB cumulative reward



(b) UCB prices over time



(c) MW cumulative reward



(d) MW prices over time

At $\beta = 4$, MW performs better than UCB because the learner's pricing disadvantage relative to the competitor is smaller. In the UCB case, the learner stays well above the competitor for most of the horizon, so the resulting positive gap $p_t - q_t$ causes a very strong drop in learner-winning probability. In the MW case, the learner is still often above the competitor, but by less. Since customer choice is extremely sensitive to price differences, this smaller gap matters a lot, as it leads to a less severe reduction in learner-winning probability and therefore to a higher cumulative reward than in UCB, making MW more robust in this environment.

UCB Learner: 3049.859 ± 66.855
MW Learner: 3604.559 ± 192.505

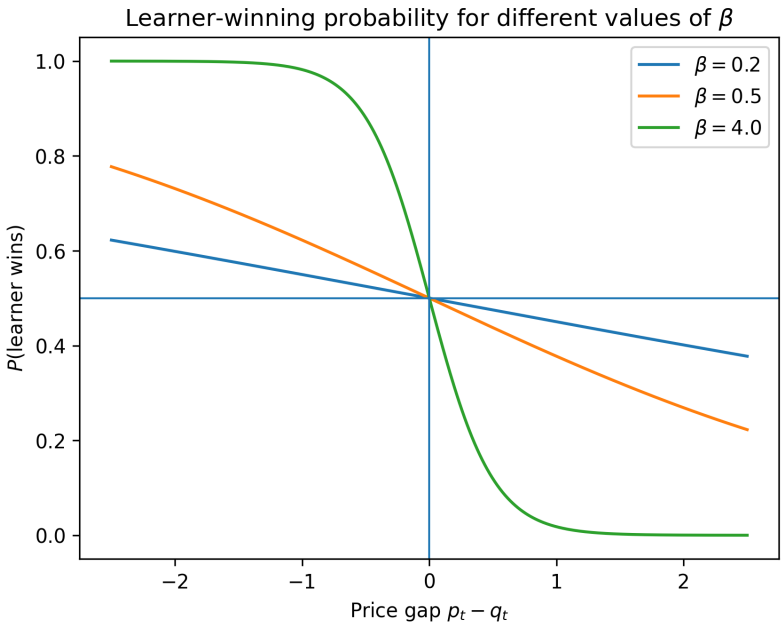


Figure 4: Learner winning probability plot

Fixed prices for the chosen seed

$$\mathbf{p}_{\text{seed}} = \begin{bmatrix} 1.2452 \\ 1.5654 \\ 3.1742 \\ 3.7904 \\ 3.9192 \end{bmatrix}$$